

EE 210L: Digital Circuits and Logic Design Lab

Lab 8: Combinational Logic in Verilog

Name: _____

Date: _____

Partner(s): _____

Section: _____

Paste your source code and waveform captures in the boxes provided. Simulation only; no hardware this week.

Part 5.1: Majority Voter



Figure 1: majority.v source.

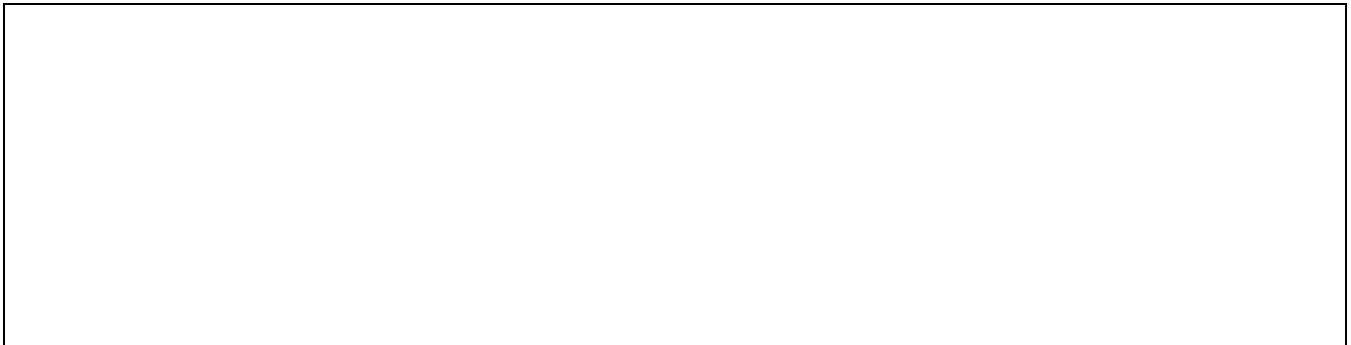
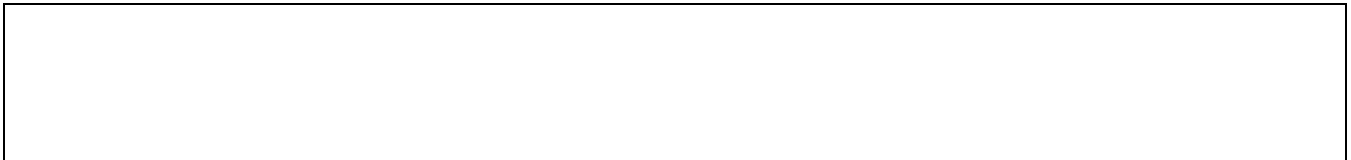


Figure 2: Simulation waveform, all eight cases.

Testbench console output:



Part 5.2: Break It on Purpose

With the a & c term removed, the testbench printed:



Which input case(s) failed?

How would you have found this on the waveform alone?

Part 5.3: Two-Bit Comparator

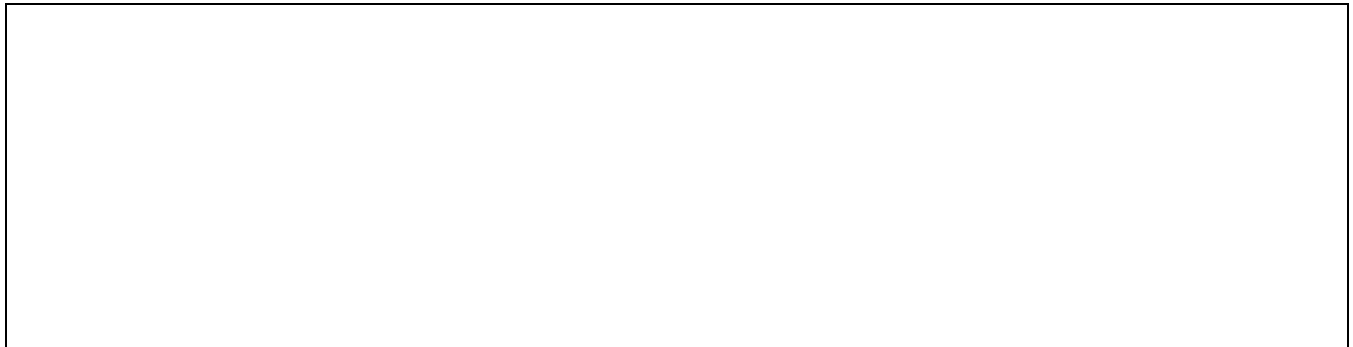


Figure 3: compare2.v source.



Figure 4: Simulation waveform, all 16 cases.

All 16 cases passed? _____ Did you add the “exactly one output high” check? ____

What class of bug does that extra check catch that testing gt, eq, and lt separately would miss?

Part 5.4: Inferred Latch

Copy the synthesis warning verbatim.



After assigning a default, was the warning gone?

Explain why the original code required a latch.

Part 5.5: Four-Bit Adder

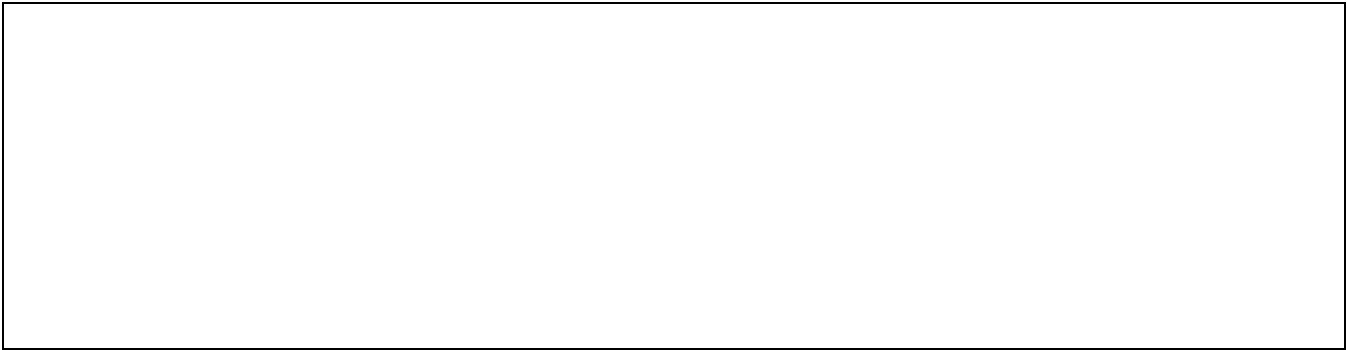


Figure 5: adder4.v and testbench source.

Cases verified	Result (pass / fail)	Approx. time to write

Compare this with wiring and measuring the 74LS283 in Lab 5. How many input combinations did you check by hand there, versus here?

Discussion and Conclusions

Address at least: what "Verilog describes hardware, it does not execute" means in practice, why a self-checking testbench beats reading waveforms, and how accidental latches happen.